

1.) $U = \text{All people}$

$\Rightarrow \forall x$
 $\wedge \exists x$

$S(x)$: x is a student
 $I(x)$: x is intelligent
 $M(x)$: x likes music

a) $\forall x (S(x) \Rightarrow I(x))$

b) $\exists x (S(x) \wedge I(x) \wedge M(x))$

c) $\forall x (M(x) \Rightarrow \neg (I(x) \wedge S(x)))$

d) $\forall x (M(x) \Rightarrow (S(x) \wedge I(x)))$

1- (8 points) Assume the universe/domain consists of all people.
Translate following statements into logic.

$S(x)$: x is a student

$I(x)$: x is intelligent

$M(x)$: x likes music

(a) all students are intelligent

(b) Some intelligent students like music

(c) Everyone that likes music is not an intelligent student

(d) Only intelligent students like music.

Q2, Q3, Q4: You must write the name of each equivalence or rule that is used.

2. Using logical equivalences and inference rules, prove that if

$\forall x (P(x) \wedge Q(x))$ then $\forall x P(x) \wedge \forall x Q(x)$

Proof of FE9: We are to prove that $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \Leftrightarrow (\forall x)(\alpha(x) \wedge \beta(x))$ is valid.

Given any interpretation I . Let U be the universal of discourse in I .

$(\forall x)\alpha(x) \wedge (\forall x)\beta(x)$ is evaluated to true

iff $(\forall x)\alpha(x)$ is evaluated to true and $(\forall x)\beta(x)$ is evaluated to true (Definition of $\wedge$)

iff $\alpha(x)$ is evaluated to true for every x in U and $\beta(x)$ is evaluated to true for every x in U (Definition of $(\forall x)F$)

iff $(\alpha(x) \wedge \beta(x))$ is evaluated to true for every x in U (Definition of $\wedge$)

iff $(\forall x)(\alpha(x) \wedge \beta(x))$ is evaluated to true. (Definition of $(\forall x)F$)

Hence, $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \Leftrightarrow (\forall x)(\alpha(x) \wedge \beta(x))$ is evaluated to true under any interpretation.

i.e. $(\forall x)\alpha(x) \wedge (\forall x)\beta(x) \equiv (\forall x)(\alpha(x) \wedge \beta(x))$.

↑ Its also a lemma equivalence

1. $\forall x (P(x) \wedge Q(x))$

2. $P(x) \wedge Q(x)$

3. $P(x)$

4. $Q(x)$

5. $(\forall x) P(x)$

6. $(\forall x) Q(x)$

7. $(\forall x) P(x) \wedge (\forall x) Q(x)$

Γ universal instantiation

UI x : variable

2, simp.

2, simp.

3, Gen.

4, Gen.

5, 6 Conjunction

3. Using logical equivalences and inference rules prove that if $\forall x (L(x) \Rightarrow R(x))$

and $\exists x (P(x) \wedge \neg R(x))$ then $\neg \forall x (P(x) \Rightarrow L(x))$

1. $\forall x (L(x) \Rightarrow R(x))$

2. $\exists x (P(x) \wedge \neg R(x))$

3. $P(c) \wedge \neg R(c)$

4. $P(c)$

5. $\neg R(c)$

6. $L(c) \Rightarrow R(c)$

7. $\neg L(c)$

8. $P(c) \wedge \neg L(c)$

9. $\exists x (P(x) \wedge \neg L(x))$

10. $\neg \neg \exists x (P(x) \wedge \neg L(x))$

11. $\neg \forall x \neg (P(x) \wedge \neg L(x))$

12. $\neg \forall x (\neg P(x) \vee L(x))$

13. $\neg \forall x (P(x) \Rightarrow L(x))$

Γ

Γ

2. EI c : constant

3. simp.

3. simp

1. VI

5, 6 m.t

4, 7 conj

8. E.G

9. double neg.

10. de morgan

11. de morgan

12. implication

4. Given the hypothesis $\exists x (A(x) \wedge \neg B(x))$ and $\forall x (A(x) \Rightarrow P(x))$

prove that $\exists x (P(x) \wedge \neg B(x))$

1. $\exists x (A(x) \wedge \neg B(x))$
2. $\forall x (A(x) \Rightarrow P(x))$
3. $A(c) \wedge \neg B(c)$
4. $A(c)$
5. $\neg B(c)$
6. $A(c) \Rightarrow P(c)$
7. $P(c)$
8. $P(c) \wedge \neg B(c)$
9. $\exists x (P(x) \wedge \neg B(x))$

- $\square$
 $\square$
1. E.I c: constant
 3. simp
 3. simp
 2. U.I c: constant
 - 4, 6. modus ponens
 - 5, 7. conjunction
 8. E.G
- allowed b/c we used E.I first